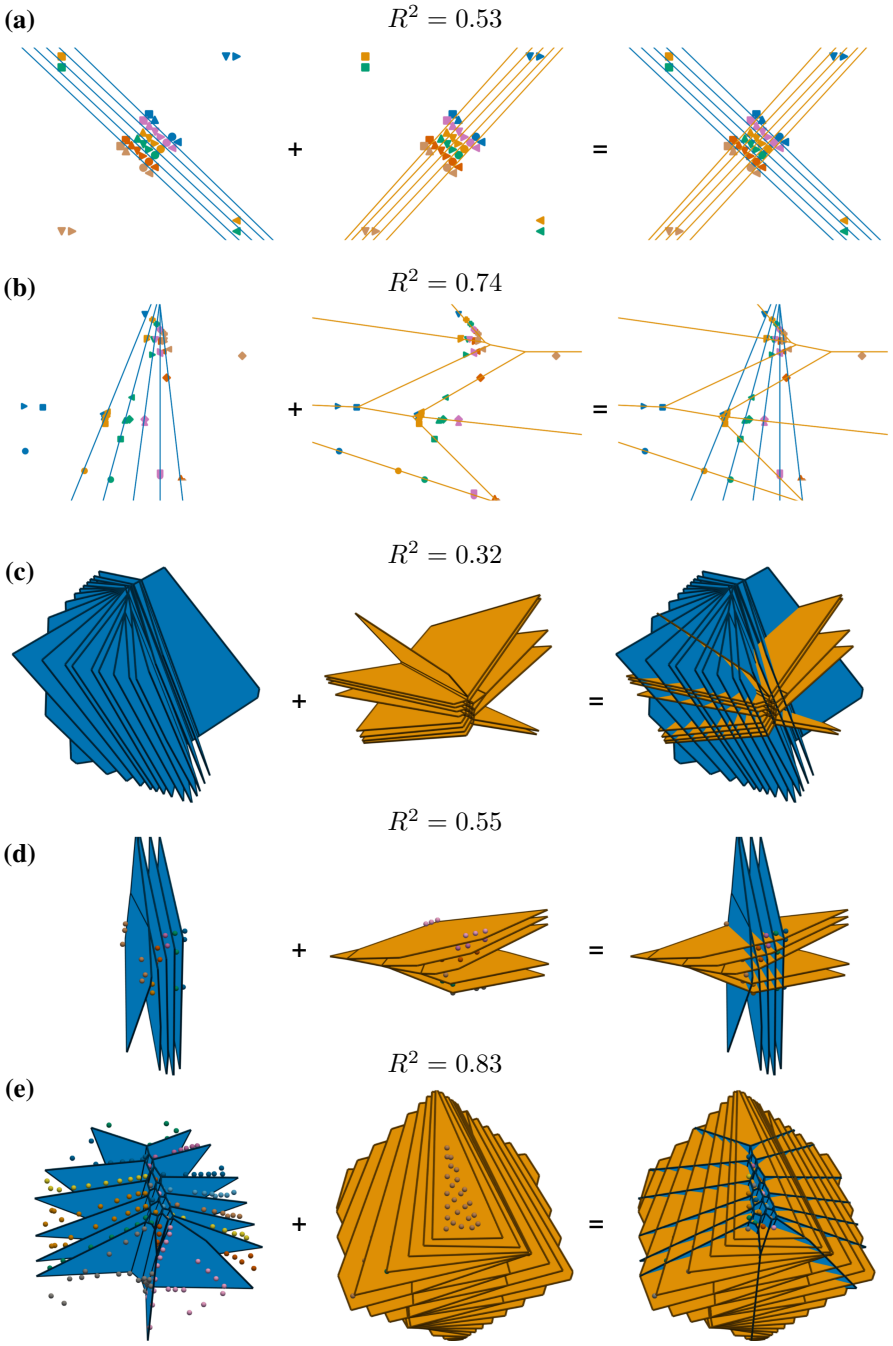

Figure 21. **Counterexamples of linear factorization under perfect classification**. The two concepts are linearly separable, but the representation cannot be decomposed into a sum of two concept components. Each subfigure shows the embedding space overlaid with three columns of argmax regions: the first column shows $\mathcal{R}_{0,i}, i \in [n]$ (shown in blue), the regions where the first concept values are predicted; the second column shows $\mathcal{R}_{1,j}, j \in [n]$ (shown in orange), the regions where the second concept values are predicted; and the third column shows $\mathcal{R}_{0,i} \cap \mathcal{R}_{1,j}, i, j \in [n]$, the joint argmax regions where specific combinations of concept values are predicted. **(a), (b)** show embeddings for two concepts (color and shape) in $\mathbb{R}^2$ ($k = d = 2$). **(d), (e)** show embedding points colored by the first concept value, all for two concepts in $\mathbb{R}^3$ ($k = 2, d = 3$). See text for details.

Table 2. **Datasets used in our main experiments.** Here k is the number of concepts, n_i is the number of values for concept i, and N is the number of samples after the preprocessing described in the Notes column.

Dataset	k	Values per concept (n_i)	N	Notes (main-experiment preprocessing)
PUG-Animal	4	`background(64), character(68), scale(3), texture(3)`	39,168	We fix `camera-yaw=0`, drop the default character texture, and remove the `Goldfish` character.
dSprites	6	`color(10), shape(3), scale(6), orient(10), posX(10), posY(10)`	180,000	We keep orientations in $[0^\circ, 90^\circ)$ (10 bins), downsample `posX/posY` to 10 bins each, and add 10 colors via on-the-fly coloring.
MPI3D	7	`obj-color(6), obj-shape(6), obj-size(2), cam-height(3), bg-color(3), horiz(10), vert(10)`	64,800	We use the real-world complex variant (original factor sizes `[6,6,2,3,3,40,40]`) and downsample the horizontal/vertical axes to 10 bins each.

Table 3. **Vision models used in our experiments.**

Family	Model identifier	Pretraining tag	Params (M)	Hidden
`openaiclip`				
clip	ViT-B/32	openai	151.3M	512
clip	ViT-L/14	openai	427.6M	768
openclip				
`metaclip2`				
openclip	ViT-H/14 (MetaCLIP2 Worldwide, 378px)	metaclip2_worldwide	1859.4M	1024
openclip	ViT-H/14 (MetaCLIP2 Worldwide, QuickGELU)	metaclip2_worldwide	1858.8M	1024
openclip	ViT-G/14 (bigG, MetaCLIP2 Worldwide)	metaclip2_worldwide	3630.4M	1280
openclip	ViT-G/14 (bigG, MetaCLIP2 Worldwide, 378px)	metaclip2_worldwide	3631.2M	1280
`siglip2`				
openclip	SigLIP2-B/16 (224)	webli	375.2M	768
openclip	SigLIP2-B/16 (256)	webli	375.2M	768
openclip	SigLIP2-L/16 (256)	webli	881.5M	1024
openclip	SigLIP2-L/16 (384)	webli	881.9M	1024
openclip	SigLIP2-SO400M/16 (256)	webli	1135.7M	1152
`siglip`				
openclip	SigLIP-B/16 (224)	webli	203.2M	768
openclip	SigLIP-B/16 (256)	webli	203.2M	768
openclip	SigLIP-L/16 (256)	webli	652.2M	1024
`openclip`				
openclip	ViT-B/16	laion400m_e32	149.6M	512
openclip	ViT-B/32	laion400m_e32	151.3M	512
openclip	ViT-L/14	laion2b_s32b_b82k	427.6M	768
dino				
`dinov3`				
dino	DINOv3 ViT-S/16	timm_default	21.6M	384
dino	DINOv3 ViT-B/16	timm_default	85.6M	768
dino	DINOv3 ViT-L/16	timm_default	303.1M	1024
`dinov2`				
dino	DINOv2 ViT-S/14	timm_default	22.1M	384
dino	DINOv2 ViT-B/14	timm_default	86.6M	768
dino	DINOv2 ViT-L/14	timm_default	304.4M	1024
`dinov1`				
dino	DINO ViT-S/16, 224px	timm_default	21.7M	384
dino	DINO ViT-B/16, 224px	timm_default	85.8M	768
timm				
`metaclip`				
timm	ViT-B/16 CLIP (MetaCLIP 2.5B, 224px)	timm_default	86.2M	768
timm	ViT-B/32 CLIP (MetaCLIP 2.5B, 224px)	timm_default	87.8M	768
timm	ViT-L/14 CLIP (MetaCLIP 2.5B, 224px)	timm_default	304.0M	1024

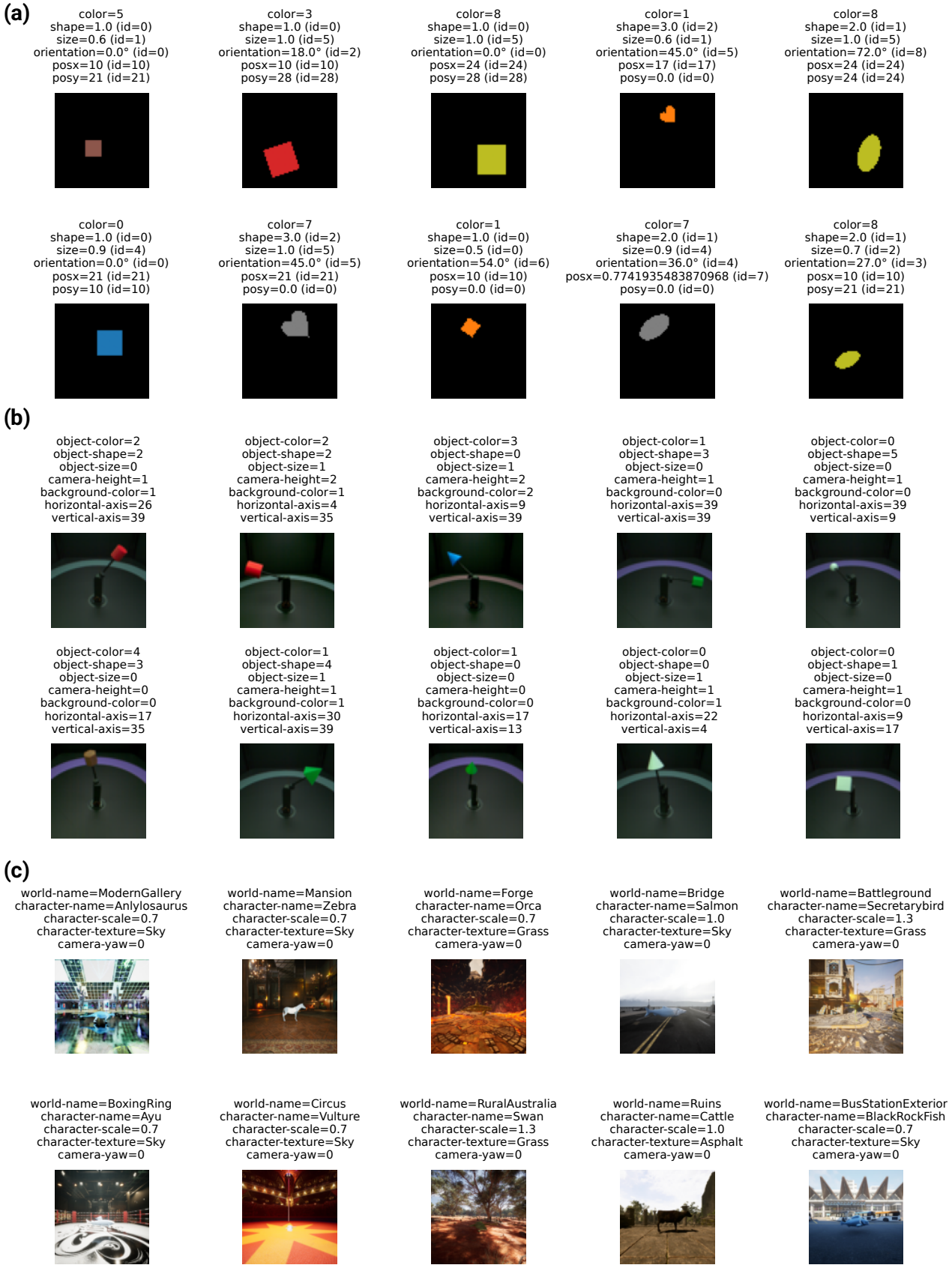

Figure 22. **Example samples from the main datasets used in our experiments. (a)** dSprites, **(b)** MPI3D, **(c)** PUG-Animal.